

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE
NAME

CLASS

TUTOR'S
NAME

CHEMISTRY

Paper 2 Structured Questions

8872/02

12 September 2011

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet.
Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Answer **two** questions on separate answer paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Section A	
1	9
2	11
3	5
4	4
5	5
6	6
Total	40

This document consists of **13** printed pages and **1** blank page

[Turn over

Section A

Answer **all** the questions in this section in the spaces provided.

- 1 Elements **A**, **B**, **C**, **D** and **E** have consecutive atomic numbers below 20. The table below shows the first four ionisation energies of elements **A**, **B**, **C**, **D** and **E**.

Element	First I.E.	Second I.E.	Third I.E.	Fourth I.E.
A	736	1450	7740	10500
B	577	1820	2740	11600
C	786	1580	3230	4360
D	1060	1900	2920	4960
E		2260	3390	4540

- (a) Identify the group to which element **A** belongs. Explain your reasoning.

Group:

Explanation:

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[2]

- (b) Suggest a possible value for the missing first ionisation energy of **E**. Explain your answer.

1st I.E. of **E**:

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[3]

- (c) Explain the difference in **second** ionisation energy between elements **A** and **B**.

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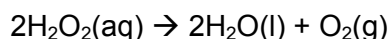
[2]

- (d) Either of the oxides of **A** or **E** is soluble in aqueous sodium hydroxide. Identify this oxide and write an equation to explain why it dissolves.

[2]

[Total: 9]

- 2 A dilute solution of hydrogen peroxide can be used to bleach hair. It decomposes slowly in aqueous solution according to the following equation.



- (a) Draw the dot-and-cross diagram to illustrate the bonding in a H_2O_2 molecule, showing the outermost shell electrons only.

[1]

- (b) A solution with an original concentration of 3.0 mol dm^{-3} was placed in a bottle contaminated with transition metal ions, which act as catalyst for the decomposition. The rate of decomposition was measured by withdrawing 10 cm^3 portions at various times and titrating with acidified $0.100 \text{ mol dm}^{-3} \text{KMnO}_4(\text{aq})$.

- (i) Write the balanced equation for the reaction between MnO_4^- and H_2O_2 .

(ii) Calculate the concentration of H_2O_2 when 30.0 cm^3 of $0.100 \text{ mol dm}^{-3} \text{ KMnO}_4$ reacts with 10.0 cm^3 of H_2O_2 .

(iii) Given that the half-life for the decomposition is 12 minutes, estimate how long the solution had been in the contaminated bottle.

[4]

(c) Explain, with reference to the *Data Booklet*, why the decomposition of H_2O_2 is an exothermic process.

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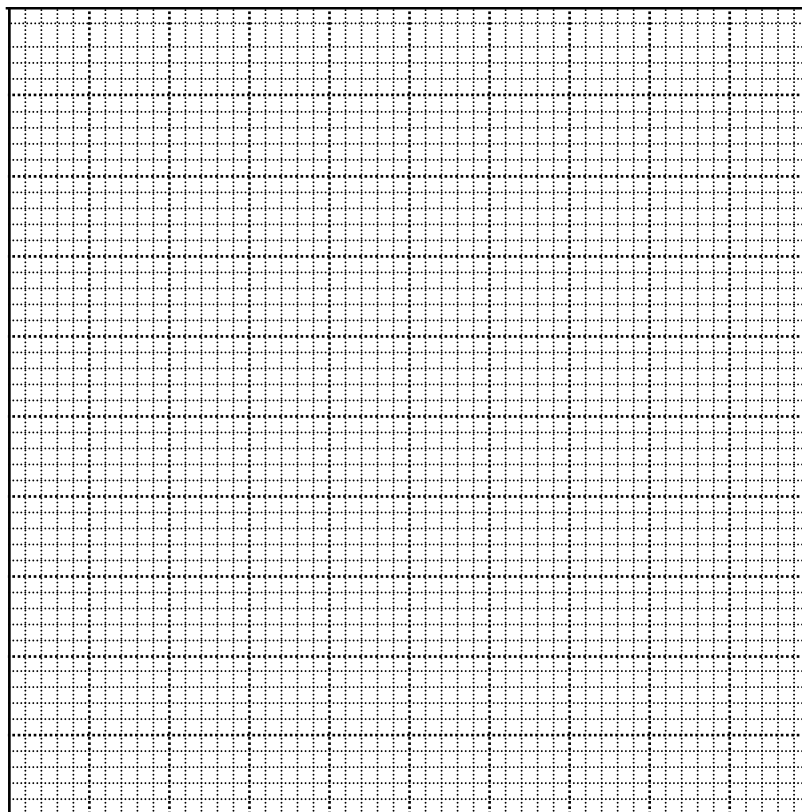
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[2]

- (d) In order to determine the rate equation for this reaction, an investigation was carried out at constant temperature. The following results were obtained.

Initial rates/ $\text{mol dm}^{-3} \text{ s}^{-1}$	16.2	47.2	78.8	124	172
$[\text{H}_2\text{O}_2]/$ mol dm^{-3}	0.09	0.25	0.44	0.67	0.90

- (i) To determine the order of reaction with respect to $[\text{H}_2\text{O}_2]$, use these data to plot a suitable graph on the grid below.



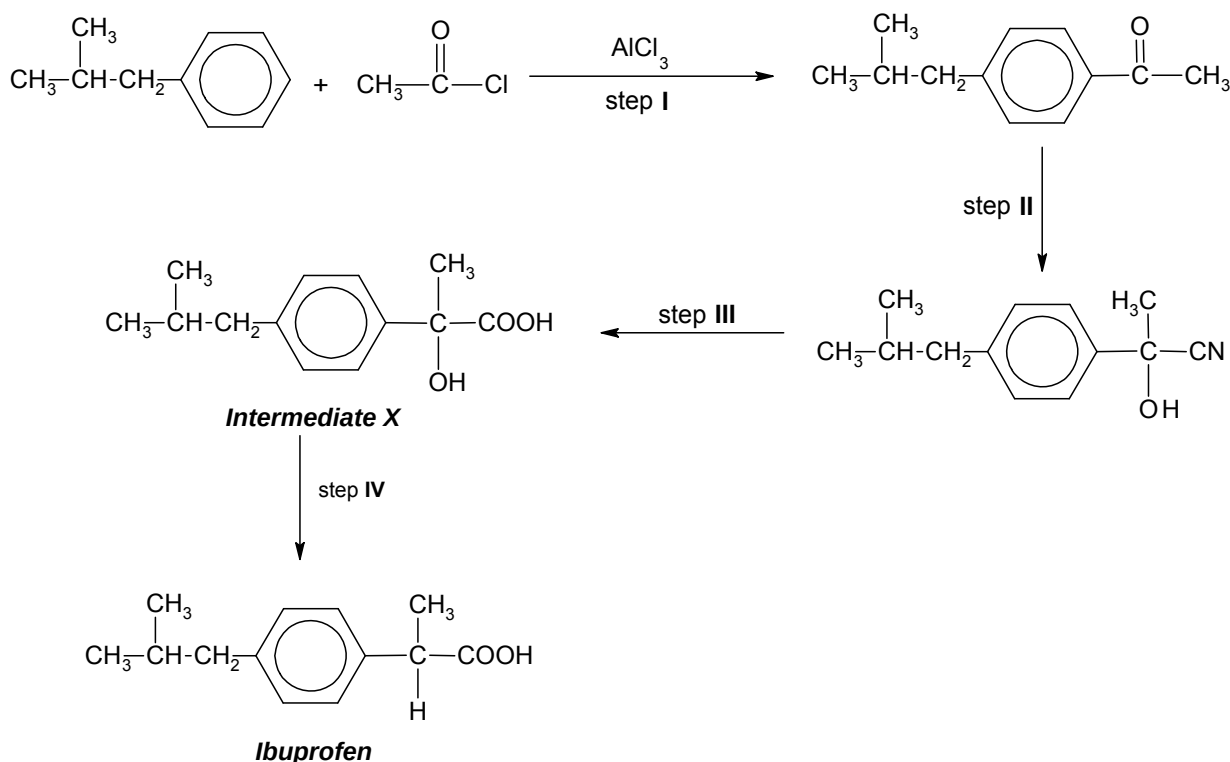
[2]

- (ii) What is the order of reaction with respect to $[\text{H}_2\text{O}_2]$?
Explain your answer.

[2]

[Total:11]

- 3 Ibuprofen is an anti-inflammatory drug used to treat fever and arthritis. The following scheme shows the synthesis of Ibuprofen from iso-butylbenzene.



- (a) Suggest the type of reaction which occurs in step I and IV.

Step I:

Step IV:

[2]

- (b) Suggest the reagents and conditions needed for step II and III.

Step	Reagent and condition
II	
III	

[2]

- (c) Draw the structural formula of the product formed when 1 mole of intermediate X (product of step III) reacts with 1 mole of methanol in the presence of concentrated sulfuric acid, under heated conditions.

[1]

[Total: 5]

- 4 Although fossil fuels have been the main energy source in the world, alcohols are becoming of increasing interest in the fuel industry as they are a renewable source of energy. Ethanol, for example, can be synthesized from the fermentation of agricultural feedstock such as corn and sugarcane.
- (a) To determine the fuel efficiency of ethanol, a chemist conducted the following experiment. 1000 cm³ of water was heated up by complete combustion of 5.0 g of ethanol and the temperature rise was measured.

Given the following data, calculate the temperature rise in the experiment. You may assume the experimental process has 80% efficiency.

	Ethanol
$\Delta H_c / \text{kJ mol}^{-1}$	-1370
M_r	46.1
Density / gcm⁻³	0.789

[3]

- (b) Ethanol is considered a “greener” fuel compared to diesel (long chain hydrocarbons). By considering the environmental impact of using diesel, suggest a reason why this is so.

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[1]

[Total: 4]

- 5 (a) An aromatic organic compound **K** has the molecular formula C_8H_8O . **K** gives a orange crystalline solid with 2,4-dinitrophenylhydrazine but does **not** react with Fehling's solution.

Suggest 2 possible identities of **K**. The 2 structures should contain 2 different functional groups.

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[2]

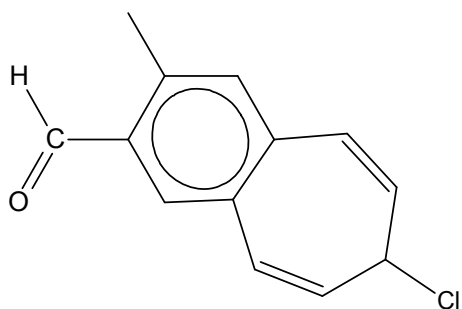
- (b) Another organic compound **L**, $C_6H_{12}O_2$, reacts with hot aqueous sodium hydroxide to form 2 products **M** and **N**. **M** has the molecular formula $C_4H_{10}O$ and it produces a yellow crystalline solid with aqueous alkaline iodine.

Identify **L**, **M** and **N** and draw the displayed formulae.

L	M
N	

[3]
[Total: 5]

- 6(a)** In the spaces provided below, draw the organic products formed when compound **J** is reacted with each of the following reagents.

Compound **J**

(i) Heat with ethanolic ammonia in a sealed tube	
(ii) Acidified potassium manganate(VII)	

[4]

- (b)** A white precipitate is observed when Compound **J** is heated under reflux with ethanolic silver nitrate.

State and explain how the rate of this reaction changes when the chloro group in compound **J** is replaced by a bromo group.

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[2]

[Total:6]

[Turn Over]

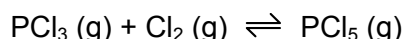
Section B

Answer **two** questions from this section on separate answer paper.

- 7(a)** Define the term *mole*.
Why is the phrase “the mass of one mole of oxygen” ambiguous? [3]
- (b)** By using the data in this question, estimate how much oxygen a person requires on average in a life-time and how far a motor car can go using the same amount of oxygen.
The typical daily food requirement of a person can be considered to be 1.2 kg of carbohydrate. The person obtains energy by the oxidation of the carbohydrate, which can be represented by the formula $(\text{CH}_2\text{O})_n$.
- (i) Construct an equation for the complete oxidation of the carbohydrate, $(\text{CH}_2\text{O})_n$.
- (ii) Given that the *empirical relative formula mass* of the carbohydrate is 30, calculate the number of moles of oxygen required by the person each day.
- (iii) Hence determine how many moles of oxygen a person require in a life-time of 70 years. [3]
- (c)** A typical motor car requires 6 dm^3 of octane, C_8H_{18} , to travel 100 km.
- (i) Construct an equation for the complete combustion of octane.
- (ii) The density of octane is 0.7 g cm^{-3} . Calculate how many moles of oxygen would be required for a typical car using octane to travel 100 km.
- (iii) From your answer in **b(iii)**, calculate how many kilometers the car can travel using the same amount of oxygen a person uses in a life-time. [4]
- (d)** Due to worldwide decline in petroleum production, scientists are trying to come up with more energy efficient octane fuel. In one experiment, 10 cm^3 of a hydrocarbon C_8H_y , were allowed to react with an excess of oxygen at 150°C and 1 atm. There was an expansion of 30 cm^3 . Deduce the value of y . [4]
- (e)** Dinitrogen pentoxide, N_2O_5 , can be produced by the following reaction sequence.
- I $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}(\text{g}) \quad \Delta H = +180 \text{ kJ mol}^{-1}$
- II $\text{NO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{NO}_2(\text{g}) \quad \Delta H = -57 \text{ kJ mol}^{-1}$
- III $2\text{NO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{N}_2\text{O}_5(\text{g}) \quad \Delta H = -55 \text{ kJ mol}^{-1}$
- (i) Explain why reaction I occurs readily in car engines, but not in the atmosphere.
- (ii) What is meant by *the standard enthalpy change of formation* of a compound?
- (iii) Use the data given to calculate the enthalpy change of formation of nitrogen dioxide.
- (iv) Hence, calculate the enthalpy change of formation of dinitrogen pentoxide. [6]

[Total:20]

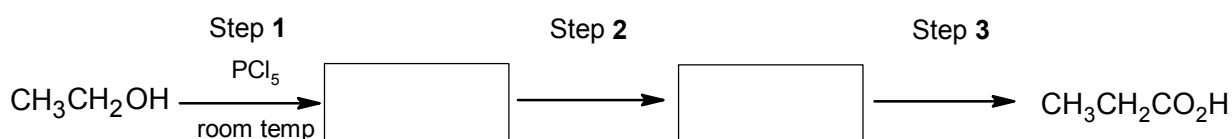
- 8(a)** Phosphorous pentachloride, PCl_5 is one of the most important phosphorus chlorides as it is used widely as a chlorinating agent. When PCl_3 and Cl_2 was placed in a 4 dm^3 reaction vessel and heated to 250°C , it was found that $1.68 \times 10^{-4} \text{ mol}$ of PCl_5 , 0.0520 mol of PCl_3 and 0.0156 mol of Cl_2 was present in the following equilibrium.



- (i) Write the K_C expression for this equilibrium.
- (ii) Calculate the K_C value at 250°C and state the units.
- (iii) Explain, using Le Chatelier's Principle, how the position of equilibrium and K_C value changes when the size of the reaction vessel is increased to 10 dm^3 .
- (iv) When the temperature decreases, more PCl_5 was formed. Suggest with reason if the forward reaction is exothermic or endothermic. [8]

- (b)** PCl_5 reacts readily in water to produce 2 mol of HCl which dissolved in water to give a solution of $\text{pH} = 1.7$

- (i) Name the reaction and write a balanced equation for the reaction.
- (i) PCl_5 is an important reactant in the production of propanoic acid, $\text{C}_3\text{H}_6\text{O}_2$, from ethanol as seen from the scheme below. Name the type of reaction in Step 1.



- (iii) Give the structures of the product for Steps 1 and 2.
- (iv) State the reagents and conditions for Steps 2 to 3 [7]

- (c)** Organic acids are usually weak acids. A 25 cm^3 sample of propanoic acid produced in **(b)** was found to weigh 0.0370 g and has a pH value of 3.5

- (i) Show, with explanation, that propanoic acid is also a weak acid.
- (ii) When half the amount of the propanoic acid reacted with equal amount of sodium hydroxide, it was found that the resulting solution pH did not change much when a small amount of acid or alkali was added to it. With the aid of equations, explain why the above observation. [5]

[Total:20]

9(a) Some physical properties of the oxides of elements **M**, **Q**, **X** and **Y** are given below:

Formula of oxide	Melting point / °C	Appearance at room temperature	Conductivity in the molten state
MO₂	1700	White solid	None
Q₂O	0	Colourless liquid	None
XO	2850	White solid	Good
YO₂	−73	Colourless gas	None

For **each** of the four oxides, what are their structures, bonding and particles present?

[4]

(b) For the reaction $2\text{HI (g)} \rightleftharpoons \text{H}_2\text{(g)} + \text{I}_2\text{(g)}$

- The enthalpy change is $+9.6 \text{ kJ mol}^{-1}$, the activation energy for the forward reaction is $+182.8 \text{ kJ mol}^{-1}$. Calculate the activation energy of the backward reaction.
- In the presence of a gold catalyst, the activation energy for the backward reaction is $+85.3 \text{ kJ mol}^{-1}$. Calculate the activation energy of the forward reaction of the catalysed reaction.
- Sketch the energy profile for the reaction, label all the values given and calculated in (i) and (ii).
- Explain, with the aid of a sketch of the Boltzmann distribution, how the presence of the gold catalyst increases the rate of the decomposition.

[8]

(c) “Cracking” in the petroleum industry refers to reactions which convert abundant long chain molecules, which have limited uses to short chain, economically useful fractions. Commonly long chain alkanes are broken up into a mixture that could consist of smaller alkanes, alkenes and or hydrogen gas.

The hydrocarbon, $\text{C}_{10}\text{H}_{22}$ was subjected to cracking and 3 non-cyclic products were obtained.

A had a mass of 58 and did not decolourise bromine in carbon tetrachloride.

B had a mass of 56 and decolourised bromine in carbon tetrachloride.

- Give a possible structure for **A**.
- Give 2 possible structures for **B**.
- Using only molecular formula write a balanced equation for the cracking process.

[4]

- (d) Propan-2-ol is found in nail polish while propane-1,2-diol is a component used in softening lotion. These two alcohols are synthesized from propene.
- (i) State the reagent and conditions for the formation of propan-2-ol from propene.
 - (ii) State the reagent and conditions for the formation of propane-1,2-diol from propene.
 - (iii) Show the structure of the product formed when propan-2-ol is heated under reflux with acidified potassium manganate(VII).
 - (iv) Show the structure of the product formed when propan-1,2-diol is heated under reflux with acidified potassium manganate(VII).

[4]

[Total: 20]

